

Philipp S. Arndt, Ph.D. | Curriculum Vitae

Data Scientist, Satellite Imagery & Data Fusion, Research & Innovation Team, Global Fishing Watch
San Diego, CA, USA • +1 858-250-9627 • psarndt93@gmail.com • flphilipp.github.io
ORCID: 0000-0002-4759-5166 • Google Scholar • GitHub: flphilipp • LinkedIn: philipparndt

Research Profile

Earth scientist and machine-learning researcher who builds automated, scalable methods that turn large volumes of satellite data into reliable, openly available information. At Global Fishing Watch, develops AI models that detect, classify, and identify vessels in optical and radar satellite imagery to support transparent monitoring of fishing and other human activity at sea. As a NASA FINESST Future Investigator at Scripps Institution of Oceanography, created algorithms that automatically detect meltwater lakes and measure their depths in NASA ICESat-2 laser-altimetry photon data across the Greenland and Antarctic ice sheets, strengthening the observational basis for projections of ice-sheet mass loss and sea-level rise.

Current focus: AI methods that fuse multi-sensor satellite imagery (Sentinel-2, PlanetScope, Sentinel-1, NISAR) with vessel-tracking (AIS, VMS) data to detect, classify, and identify non-broadcasting (“dark”) vessels at global scale, flag suspicious behavior, and map activities such as illegal fishing, transshipment, bottom trawling, sand dredging, and small-scale fishing in coastal waters.

Research Highlights

- **Federal research support:** NASA FINESST award (\$135,000, 2020–2024) for doctoral research combining ICESat-2 and machine learning to study Antarctic ice shelves; ice-sheet-scale computing on the NSF-supported OSG Open Science Pool; fieldwork with NSF-funded projects in Antarctica (U.S. Antarctic Program, 2019) and Alaska (2022).
- **Publications and peer review:** Four peer-reviewed journal articles (*Geophysical Research Letters*, *The Cryosphere*), including a first-author methods paper, and a doctoral dissertation; 175 citations on Google Scholar, h-index 4 (September 2026); peer reviewer for *The Cryosphere*.
- **Use of work by others:** Code and datasets openly released with DOIs; ICESat-2 lake-depth code used by independent research groups (e.g., Lutz et al., 2024, *The Cryosphere*); visualization tutorial published as a chapter of Project Pythia’s open-source *ICESat-2 Cookbook* (2026).
- **Global maritime monitoring:** Contributor to Global Fishing Watch’s July 2025 global release of Sentinel-2 vessel detections, which capture about three times more vessels than radar and cover all coastal waters; now developing classification models for vessel detections in Planet’s 3 m imagery.
- **Dissemination and media:** 16 presentations and tutorials as presenting author (incl. AGU Fall Meetings, NASA ICESat-2 Science Team Meetings, WAIS Workshops, and the ESA–NASA Cryo2ice Symposium) and 4 co-authored conference abstracts (incl. two invited AGU talks and the 2026 Our Ocean Conference); research covered by UC San Diego Today, Columbia Climate School’s *State of the Planet*, EarthSky, ScienceAlert, SciTechDaily, and The Weather Network.

Education

Scripps Institution of Oceanography, UC San Diego: *Ph.D. Earth Sciences*, Dec 2024

Dissertation: “Using High-Resolution Satellite Observations to Better Understand Surface Melt Processes on the Greenland and Antarctic Ice Sheets”

Advisor: Prof. Helen Amanda Fricker • Public dissertation defense: 16 September 2024 • La Jolla, CA, USA

Scripps Institution of Oceanography, UC San Diego: *M.Sc. Earth Sciences*, Jan 2020
La Jolla, CA, USA (cumulative GPA: 3.97/4.00)

Chalmers University of Technology: *M.Sc. Engineering Physics (Complex Adaptive Systems)*, Jun 2018
Thesis: “Detection of Multi-Level Hierarchies in Multi-View Cancer Networks, with Applications to Glioblastoma Multiforme”
Gothenburg, Sweden (cumulative GPA: 4.55/5.00)

Yale University: *B.A. Economics & Mathematics*, May 2016
New Haven, CT, USA (cumulative GPA: 3.66/4.00)

University of Alaska Fairbanks: *International Summer School in Glaciology*, Jun 2022
McCarthy, AK, USA

Stanford University: *Summer Studies*, 2011
Stanford, CA, USA (GPA: 3.89)

Research & Professional Experience

Global Fishing Watch | Data Scientist, Satellite Imagery & Data Fusion (Research & Innovation Team)
Jan 2025 – present, San Diego, CA, USA (remote)

International nonprofit that publishes open data on human activity at sea. Builds machine-learning systems that detect vessels in optical (Sentinel-2, PlanetScope) and radar (Sentinel-1, NISAR) satellite imagery and determine their type, activity, and identity. The team collaborates closely with partner nonprofits SkyTruth (monitoring offshore infrastructure) and the Allen Institute for AI (Ai2; maritime foundation models and vessel detection in NASA–ISRO NISAR radar data).

- **Global Sentinel-2 vessel detections (released July 2025):** Rebuilt the database of environmental and contextual features (e.g., bathymetry, seabed type, ocean and weather conditions, distance to shore and port, AIS vessel density, socioeconomic indicators) that the vessel-classification models combine with image embeddings. The open dataset captures about three times more vessels than radar, including some under 10 m, across all coastal waters.
- **Planet data release:** Helps develop the vessel-classification models for GFW’s release of detections in Planet’s 3 m PlanetScope imagery, which brings small coastal vessels (down to ~5 m) into view. Manages image annotation for training data (wrote and shared labeling tools; trained and manages an external labeling contractor firm) and builds BigQuery feature pipelines across hundreds of millions of detections.
- **Dark vessels and new activity types:** Supporting a manuscript in preparation on retrieving the identities of large (>50 m) vessels that do not broadcast AIS; worked on sand-dredging classification in Sentinel-2 and PlanetScope imagery and onboarded collaborators from UC Santa Barbara’s emLab to the effort.
- **Bottom trawling in Australia (for the Munderoo Foundation):** Co-led a project mapping bottom trawling across Australia’s Exclusive Economic Zone by combining AIS data with Sentinel-2 detections, including how much occurs inside versus outside Fisheries Restricted Areas; deliverables: maps, a report, and a presentation to the Foundation.

Scripps Institution of Oceanography | Graduate Researcher & NASA FINESST Future Investigator
Sep 2018 – Dec 2024, UC San Diego, La Jolla, CA, USA Advisor: Prof. Helen Amanda Fricker
Developed automated methods to measure meltwater on the Greenland and Antarctic ice sheets with NASA’s ICESat-2 laser altimeter and multispectral satellite imagery, to better understand how surface melt affects ice-shelf stability, ice flow, and sea-level rise.

- **Automated lake-depth retrieval at ice-sheet scale:** Developed FLUID–SuRRF, a fully automated framework that detects meltwater lakes in ICESat-2 photon data and measures their depths (mean abso-

lute error <0.27 m; Arndt & Fricker, 2024, *The Cryosphere*). Scaled it with distributed high-throughput computing on the NSF-supported Open Science Pool to process 138 TB of data (~ 765 billion photons) from five melt seasons on both ice sheets, yielding 10,954 lake-depth segments up to 33 m deep.

- **Machine learning for meltwater volume:** Trained a random-forest model on the Greenland ICESat-2 depths to estimate lake depth and volume from Sentinel-2 imagery; it reaches ~ 0.5 m depth accuracy against in situ sonar, cuts volume errors by over 60% compared with traditional radiative-transfer methods, and generalizes to Antarctica. Manuscript in preparation with H. A. Fricker for the *Annals of Glaciology* special collection on AI and machine-learning methods, with a planned update to eight years of ICESat-2 data (2019–2026).
- **Open tools and collaborative studies:** Wrote the open-source ICESat-2 depth-retrieval code for Fricker et al. (2021, *GRL*), since used by independent research groups (e.g., Lutz et al., 2024, *The Cryosphere*); contributed analysis, code, and the widely republished water-volume graphic to Warner et al. (2021, *GRL*) on the sudden drainage of an Antarctic ice-shelf lake; compiled an inventory of through-ice-shelf lake drainages across Antarctica.
- **Fieldwork and mentoring:** Designed and led a ski-based campaign to ground-validate ICESat-2 snow depths in the San Bernardino Mountains (2023); mentored three undergraduate research projects and was a teaching assistant for two courses.

Fraunhofer-Chalmers Centre for Industrial Mathematics | Contracted Student Researcher

Jan 2017 – Dec 2017, Gothenburg, Sweden Supervisor: Mats Jirstrand

- Analyzed GPS tracking data for a large truck manufacturer to gain insights into its customers' mobility patterns.
- Created a global clustering of truck-stop locations, cross-referenced the resulting “hubs” with existing infrastructure data, and trained a classifier to categorize unlabeled locations.

Yale Graduate School of Arts and Sciences | Visiting Assistant in Research

Jun 2017 – Aug 2017, New Haven, CT, USA Supervisors: Trude Storelvmo & Peter C. B. Phillips

- Estimated Earth's transient climate sensitivity from in-situ weather-station data using econometric methods; co-author of the resulting Storelvmo et al. (2018, *GRL*).
- Developed machine-learning algorithms for spatio-temporal interpolation of weather-station records based on global reanalysis data.

Potsdam Institute for Climate Impact Research | COPAN Guest Researcher

May 2015 – Aug 2015, Potsdam, Germany Supervisor: Jobst Heitzig

- Found that most of the “Great Acceleration” socio-ecological time series exhibit statistically significant break points around 1950.
- Showed that the Anthropocene could be defined as a stratigraphy-based geological epoch backed by a robust statistical analysis of its defining socioeconomic drivers.

Munich Re | Client Management and Underwriting Intern

May 2014 – Jul 2014, Munich, Germany

Risk modeling, excess-of-loss (XL) treaty pricing, solvency analysis, business strategy

Yale Office of Public Affairs & Communications | Social Media & Photography Intern

Aug 2013 – May 2015, New Haven, CT, USA

Photography, outreach strategy, and official representation of Yale University on social media

Peer-Reviewed Publications

Google Scholar (September 2026): 175 citations, h-index 4, i10-index 4
<https://scholar.google.com/citations?user=NORXhMcAAAAJ>

- [1] **Arndt, P. S.** & Fricker, H. A. (2024). A framework for automated supraglacial lake detection and depth retrieval in ICESat-2 photon data across the Greenland and Antarctic ice sheets. *The Cryosphere*, 18(11), 5173–5206. doi:10.5194/tc-18-5173-2024 [cited by 19]
- [2] Warner, R. C., Fricker, H. A., Adusumilli, S., **Arndt, P.**, Kingslake, J., & Spergel, J. J. (2021). Rapid Formation of an Ice Doline on Amery Ice Shelf, East Antarctica. *Geophysical Research Letters*, 48(14), e2020GL091095. doi:10.1029/2020GL091095 [cited by 40]
- [3] Fricker, H. A., **Arndt, P.**, Brunt, K. M., Datta, R. T., Fair, Z., Jasinski, M. F., Kingslake, J., Magruder, L. A., Moussavi, M., Pope, A., Spergel, J. J., Stoll, J. D., & Wouters, B. (2021). ICESat-2 Meltwater Depth Estimates: Application to Surface Melt on Amery Ice Shelf, East Antarctica. *Geophysical Research Letters*, 48(8), e2020GL090550. doi:10.1029/2020GL090550 [cited by 76]
- [4] Storelvmo, T., Heede, U. K., Leirvik, T., Phillips, P. C. B., **Arndt, P.**, & Wild, M. (2018). Lethargic Response to Aerosol Emissions in Current Climate Models. *Geophysical Research Letters*, 45(18), 9814–9823. doi:10.1029/2018GL078298 [cited by 34]

Doctoral Dissertation

Arndt, P. S. (2024). *Using High-Resolution Satellite Observations to Better Understand Surface Melt Processes on the Greenland and Antarctic Ice Sheets*. Ph.D. dissertation, Scripps Institution of Oceanography, University of California San Diego. <https://escholarship.org/uc/item/7hk6f0dr>

Research Grants & Fellowships

- 2020–24 **Future Investigators in NASA Earth and Space Science and Technology (FINESST)** research award, NASA Earth Science Division: \$135,000 (grant 80NSSC20K1666). Project: “Exploitation of ICESat-2’s Unique Capabilities and Machine Learning for Improved Understanding of Mass Balance Processes Across all Antarctic Ice Shelves” (Future Investigator; PI: H. A. Fricker)
- 2020–23 *Wyer Family Endowed Fellowship* for PhD stipend supplement
- 2020–23 *Katzin Fellowship* for PhD program tuition supplement
- 2018–19 *Scripps Fellowship* for outstanding incoming graduate students
- 2015 *Richter Fellowship* for climate research in Germany
- 2015 *e-fellows.net Scholarship*, awarded for academic merit
- 2013 *Larry Coben Fellowship* for academic work in Brazil

Honors & Travel Awards

- 2023 *Early-Career Researcher Travel Grant*, WAIS Workshop (Cloquet, MN)
- 2022 Selected speaker, *Graduate Student Spotlight*, NASA ICESat-2 Science Symposium & Science Team Meeting (Austin, TX)
- 2022 *Early-Career Researcher Travel Grant*, WAIS Workshop (Estes Park, CO)
- 2022 *WSL Early-Career Researcher Travel Grant*, The Cryosphere in a Changing Climate (Davos, Switzerland)
- 2021 *Scripps Departmental Travel Grant*, AGU Fall Meeting (New Orleans, LA)
- 2021 *Early-Career Researcher Travel Grant*, WAIS Workshop (Sterling, VA)
- 2019 *Antarctica Service Medal*, U.S. Department of Defense

- 2019 *Scholar-Athlete Award*, National Association of Intercollegiate Gymnastics Clubs
2019 *Academic Excellence Award*, UC San Diego Club Sports

Open Data Products & Software

Archived with DOIs on Zenodo; further repositories at <https://github.com/flphilipp>

- Arndt, P. S.** & Fricker, H. A. (2024). Source code for FLUID–SuRRF, automated supraglacial lake detection and depth retrieval in ICESat-2 photon data (incl. a Singularity container for distributed high-throughput computing). Zenodo. doi:10.5281/zenodo.10905941
- Arndt, P. S.** & Fricker, H. A. (2024). Data for “A framework for automated supraglacial lake detection and depth retrieval in ICESat-2 photon data across the Greenland and Antarctic ice sheets”: 1,249 ICESat-2 lake segments with along-track depths (HDF5). Zenodo. doi:10.5281/zenodo.10901737
- Arndt, P.** (2024). Data and code for figure reproduction for the same paper (FLUIDSuRRF–figures). Zenodo. doi:10.5281/zenodo.10901826
- Arndt, P.** (2025). ICESat2Sentinel2 (v0.1.0): open-source code for pairing ICESat-2 data with Sentinel-2 imagery. Zenodo. doi:10.5281/zenodo.14722234
- Arndt, P.**, Warner, R., Fricker, H. A., Adusumilli, S., Kingslake, J., & Spergel, J. (2021). Data and code for “Rapid Formation of an Ice Doline on Amery Ice Shelf, East Antarctica.” Zenodo. doi:10.5281/zenodo.4747428
- Arndt, P.** (2020). Code for ICESat-2 melt-lake depth retrieval comparisons on Amery Ice Shelf, for Fricker et al. (2021) (ameryMeltLakesICESat2, v1.0.0). Zenodo. doi:10.5281/zenodo.4299237
- Arndt, P.** pondpicking: interactive Jupyter tool for ICESat-2 lake-depth retrieval, used in Lutz et al. (2024, *The Cryosphere*). <https://github.com/flphilipp/pondpicking>
- Arndt, P.** & Roberts, C. (2026). Interactive Visualization with OpenAltimetry and Google Earth Engine. Chapter in *ICESat-2 Cookbook* (v1.0.0), Project Pythia. doi:10.5281/zenodo.21298648
- ICESat-2 Hackweek 2022 and 2023 websites and tutorials (co-author), eScience Institute, University of Washington. doi:10.5281/zenodo.6462479; doi:10.5281/zenodo.10519966
- Contributor (team product):** Global Fishing Watch Sentinel-2 vessel detections, a global, openly available dataset (GFW map, data portal, and monthly Zenodo releases), 2025–present.

Presentations

Presenting author is listed first; abstracts presented by a co-author are marked “co-author.”

- Wei, Z., Kroodsmas, D., Hochberg, T., Li, Y., Paolo, F., **Arndt, P.**, Tuda, P., Balinger, O., Azevedo, A., Requero, R., Piagna, L., McGhee, G., DeLand, S. & Cleary, J. (2026). Satellite Imagery Reveals Small-Scale Fishing Activity and Industrial Encroachment across Africa. *11th Our Ocean Conference (OOC11), Academia and Research Pre-Conference Symposium*, 15 June 2026, Mombasa, Kenya. [Poster P2.5; co-author; presented by D. Kroodsmas]. Abstract book: ouroceankenya.com
- Arndt, P. S.** (2024). Five Years of ICESat-2-Derived Supraglacial Lake Depths Facilitate Advancements in Meltwater Volume Estimation and Novel Insights into Hydrological Processes on the Ice Sheets. *Cryo2ice Symposium 2024* (ESA & NASA), Session 3: Ice Sheets and Ice Shelves – Hydrology & Dynamics, 23 September 2024, Harpa, Reykjavík, Iceland. [Oral]. Program: cryo2ice-symposium.org/detailed-agenda
- Arndt, P. S.** (2024). Using High-Resolution Satellite Observations to Better Understand Surface Melt Processes on the Antarctic and Greenland Ice Sheets. *Doctoral Dissertation Defense*, Scripps Institution of

- Oceanography, UC San Diego, 16 September 2024, Munk Conference Room (303), La Jolla, CA, USA. [Public Oral Defense]
- Arndt, P.** & Fricker, H. A. (2023). Supraglacial Lakes across Greenland and Antarctica – New Insights from Five Years of ICESat-2 Meltwater Depth Measurements. *NASA ICESat-2 Science Team Meeting*, 2–3 October 2023, La Jolla, CA, USA. [Oral Lightning Talk & Poster]
- Arndt, P.** & Fricker, H. A. (2023). ICESat-2 Measurements of Supraglacial Lake Depths Across the Antarctic and Greenland Ice Sheets 2018–2023. *2023 West Antarctic Ice Sheet (WAIS) Workshop*, 28 September 2023, Cloquet, MN, USA. [Oral]. Recording: youtu.be/d_7SI7gLrgE
- Arndt, P.** (2023). Interactive Data Visualization with OpenAltimetry & Contextualization of ICESat-2 Photon Data using Sentinel-2 Cloud Masking in Google Earth Engine. *ICESat-2 Hackweek 2023*, eScience Institute, University of Washington, 9 August 2023, Seattle, WA, USA. [Hybrid Tutorial]. Recording: youtu.be/HKIQqiHAWsA
- Arndt, P.**, Roberts, C. & The UCSD Alpine Club (2023). We Skied Southern California To Show that Mountain Snowpack Can Be Measured from Space. *Scripps Fellowship Luncheon*, 20 June 2023, La Jolla, CA, USA. [Poster & Oral Presentation, Panel Discussion]
- Arndt, P.**, Roberts, C. & The UCSD Alpine Club (2023). Skiing SoCal for Science – Preliminary Results from Ground-Validating ICESat-2 Satellite Laser Altimetry Data for Snow Depth Retrievals in the San Bernardino Mountains. *Scripps Polar Center Town Hall*, 1 June 2023, La Jolla, CA, USA. [Oral]
- Arndt, P. S.** & Fricker, H. A. (2022). Towards Automated Generation of Ice-Sheet Wide Supraglacial Meltwater Depth Measurements from ICESat-2 Data, Using High-Throughput Computing (abstract C35C-0895). *AGU Fall Meeting 2022*, 14 December 2022, Chicago, IL, USA. [Poster]
- Arndt, P.** & Fricker, H. A. (2022). Towards Automated Retrieval of Supraglacial Lake Depth Measurements from ICESat-2 Data Across Antarctica’s Ice Shelves. *NASA ICESat-2 Science Symposium and Science Team Meeting*, 27 October 2022, Austin, TX, USA. [Graduate Student Spotlight, Oral]
- Arndt, P.** & Fricker, H. A. (2022). Towards Automated Retrieval of Supraglacial Lake Depth Measurements from ICESat-2 Data Across Antarctica’s Ice Shelves. *2022 WAIS Workshop*, 29 September 2022, Estes Park, CO, USA. [Oral]. Recording: youtube.com/watch?v=u5-i6uNmD7o
- Arndt, P.**, Dundovic, D. & Bueler, E. (2022). Stokes modelling of glacier flow: The effects of surface mass perturbations. *International Summer School in Glaciology, Public Presentations*, 23 June 2022, McCarthy, AK, USA. [Oral]
- Arndt, P.** (2022). Interactive Data Visualization with OpenAltimetry & Google Earth Engine. *ICESat-2 Hackweek 2022*, eScience Institute, University of Washington, 23 March 2022, virtual. [Online Tutorial, co-presented with W. J. Leong]. Recording: youtube.com/watch?v=fpcBFsy6Kso
- Arndt, P. S.** & Fricker, H. A. (2021). Through-Ice-Shelf Drainage of Surface Meltwater Lakes and its Implications for Antarctic Ice Shelf Stability (abstract C45F-1072). *AGU Fall Meeting 2021*, 16 December 2021, New Orleans, LA, USA. [Poster: “We found many examples of melt lakes draining through ice shelves around Antarctica”]. doi:10.1002/essoar.10510105.1
- Gardner, A. S., Adusumilli, S., **Arndt, P. S.**, Brunt, K., Csatho, B. M., Felikson, D., Paolo, F., Fricker, H. A., Greene, C. A., Kacimi, S., Kurtz, N. T., Kwok, R., Medley, B., Neumann, T., Nilsson, J., Petty, A., Shean, D. E., Siegfried, M. & Smith, B. E. (2021). Surface Topography Observations Needed to Advance Cryosphere Science in the Coming Decades (abstract U32A-05). *AGU Fall Meeting 2021*, 15 December 2021, New Orleans, LA, USA. [Invited Union Session Talk; co-author]
- Arndt, P.**, Fricker, H. A., Warner, R., Adusumilli, S., Kingslake, J. & Spergel, J. J. (2021). Through-ice-shelf drainage of surface meltwater lakes and its implications for ice shelf stability. *2021 WAIS Workshop*, 22 September 2021, Sterling, VA, USA. [Oral]. Recording: youtube.com/watch?v=bP-

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- Warner, R., Fricker, H., Adusumilli, S., **Arndt, P.**, Kingslake, J. & Spergel, J. (2021). Rapid formation of an ice doline on Amery Ice Shelf, East Antarctica (EGU21-14157). *vEGU21: EGU General Assembly 2021*, April 2021, virtual. [Display; co-author]. doi:10.5194/egusphere-egu21-14157
- Arndt, P.** (2021). How to survey supraglacial melt lakes in Antarctica, from your home office. *Scripps Polar Center Virtual Research Town Hall*, 4 February 2021, virtual. [Lightning Talk]. Recording: youtube.com/watch?v=0kLTPv5XIRM
- Fricker, H. A., **Arndt, P. S.**, Brunt, K. M., Datta, R., Fair, Z., Jasinski, M. F., Kingslake, J., Magruder, L. A., Moussavi, M. S., Spergel, J., Stoll, J. & Wouters, B. (2020). ICESat-2 melt depth retrievals: Revisiting surface melt on Amery Ice Shelf, East Antarctica (abstract C058-01). *AGU Fall Meeting 2020*, 15 December 2020, virtual. [Invited Oral; co-author]
- Arndt, P.**, Fricker, H. A., Brunt, K. M., Datta, R. T., Fair, Z., Jasinski, M. F., Kingslake, J., Magruder, L. A., Moussavi, M., Pope, A., Spergel, J. J., Stoll, J. D., Wouters, B. & Warner, R. (2020). A new technique for measuring the depths of surface melt features on ice sheets from ICESat-2. *2020 WAIS Workshop*, 24 September 2020, virtual. [Oral]. Recording: youtube.com/watch?v=Sd04lclFsMA

Media Coverage

Coverage of Warner et al. (2021) unless noted; “visualization credited” marks articles that feature Arndt’s graphic of the lost water volume.

UC San Diego Today / Scripps Institution of Oceanography News (23 June 2021). “Scientists Track Sudden Disappearance of Antarctic Ice Shelf Lake.” [names Arndt as co-author]

State of the Planet, Columbia Climate School (24 June 2021). “Scientists Track the Sudden Disappearance of an Antarctic Ice-Shelf Lake.”

EarthSky (28 June 2021). “An Antarctic lake suddenly disappears.” [visualization credited]

ScienceAlert (28 June 2021). “Gigantic Antarctic Lake Suddenly Disappears in Monumental Vanishing Act.” [visualization credited]

SciTechDaily (29 June 2021). “Sudden Disappearance of Giant Antarctic Lake Leaves Massive Crater – 200 Billion Gallons of Water Gone.” [visualization credited; names Arndt as co-author]

The Weather Network (4 July 2021). “Disappearing lake in Antarctica leaves scientists puzzled.” [visualization credited]

UC San Diego Graduate Division. “Graduate Student Spotlight: Philipp Arndt” [profile of doctoral research and the NASA FINESST award]

Field Experience

UC San Diego Alpine Club | Field Team & Project Lead (Independent Research)

April 2023, San Bernardino National Forest, CA, USA

Ski-based ICESat-2 ground-validation campaign for snow-depth measurements on San Geronio Mountain

University of Alaska Fairbanks / National Science Foundation | Field Assistant

May 2022 – Jun 2022, McCarthy, AK, USA

Anticipating Rates of Deglaciation in Alaska: Controls on the Mass Loss and Morphology of the Debris-Covered Terminus of Kennicott Glacier, Wrangell–St. Elias National Park (NSF Award 1917536)

United States Antarctic Program / National Science Foundation | Field Researcher (Grantee)

Oct 2019 – Dec 2019, McMurdo Station & Siple Dome, Antarctica

Subglacial Antarctic Lakes Scientific Access (SALSA) project, geophysics (event number C-533): two-month deep-field deployment living in tents on the ice, including recovery of GPS instruments by small aircraft

Bavarian Academy of Sciences, Commission for Geodesy & Glaciology | Glacier Field Course
Jul 2010, Vernagtferner Glacier, Ötztal Alps, Austria

Teaching & Mentorship

University of Washington eScience ICESat-2 Hackweeks | Organizer, Tutorial Developer & Lead
Mar 2022 (virtual) & Aug 2023 (Seattle, WA, USA)

Developed and taught tutorials on data discovery and interactive visualization of ICESat-2 data with OpenAltimetry and Google Earth Engine; the tutorial was later published as a chapter of Project Pythia's open-source *ICESat-2 Cookbook* (2026)

Scripps Institution of Oceanography | Undergraduate Research Mentor (SIO 199: Independent Research)
Spring 2023 & Spring 2024, La Jolla, CA, USA

Supervised three student projects, providing research direction and technical support for data retrieval, cloud computing, and analysis:

- Morphology and Timing of a Recent Calving Event on Filchner-Ronne Ice Shelf
- Characterizing Seasonal Fluctuations in Ice-Marginal Lake Levels in Northwestern Greenland
- Grounding Zone Dynamics at the Eastern Margin of the Amery Ice Shelf

Scripps Institution of Oceanography | Teaching Assistant

SIO 173: Dynamics of the Atmosphere and Climate, Spring 2024

SIO 115: Ice in the Climate System (instructor: Helen A. Fricker), Winter 2020

Instituto Dois Irmãos | English Teacher

Jun 2013 – Aug 2013, Rocinha, Rio de Janeiro, Brazil

Trinity International School | Resident Teacher

May 2012 – Jul 2012, Tirunelveli, India

Professional Service, Leadership & Community Engagement

- Peer reviewer (anonymous referee), *The Cryosphere*, 2021; public community comments in *The Cryosphere*'s open peer-review discussions of later manuscripts
- Organizer and tutorial lead, ICESat-2 Hackweeks, University of Washington eScience Institute, 2022 & 2023
- Board Member, Allied Climbers of San Diego, 2024–present
- Founding member, instruction and outreach, UC San Diego Alpine Club, 2022–present
- Sustainability Ambassador (publicity and science communication), Chalmers Environmental Unit, 2017–18
- Student Representative, M.Sc. Complex Adaptive Systems program, Chalmers University of Technology and University of Gothenburg, 2017–18
- Artistic Gymnastics Coach (ages 6 to mid-50s) and Judge, Göteborgs Turnförening, 2016–18
- Co-Founder, President (2013–15), and Coach (2013–16), Yale Gymnastics Club
- President (2014–15) and Events Coordinator (2013–16), German Society at Yale University
- Co-Founder and President, Yale Photography Society, 2012–13
- Photography: free or heavily discounted photo shoots for small community organizations, alongside paid freelance projects and work for Yale University, 2013–16

- Artistic Gymnastics Coach (ages 6–14) and Judge, Turn- und Sportverein Fürstenfeldbruck, 2007–11

Skills

Methods: Machine learning and computer vision for satellite imagery • Sensor processing and data fusion (optical, SAR, lidar altimetry, AIS) • Satellite remote sensing • Large-scale geospatial data analysis • Distributed high-throughput computing • Training-data design and annotation management • Statistical learning and econometrics • Data visualization • Scientific writing and communication

Computing: Python • SQL (Google BigQuery, PostgreSQL/PostGIS) • PyTorch • scikit-learn • Jupyter • Git/GitHub • bash • Google Cloud Platform & AWS • Google Earth Engine • HTCondor • Singularity/Docker • Spark • MATLAB • R • C • Java • JavaScript • Stata • Mathematica • QGIS & ArcGIS • L^AT_EX • AI coding assistants (Claude Code, Gemini CLI)

Languages: German (native) • English (fluent) • French (conversational) • Portuguese (conversational)

Certifications & Licenses

- Wilderness First Responder (WFR), Wilderness Medical Associates, 2022–2028
- Single Pitch Instructor (SPI), American Mountain Guides Association, 2024–2027
- PADI Open Water Diver
- Driver's licenses: California & Germany

Selected Coursework

Graduate: Satellite Remote Sensing • Ice Sheet-Ocean Interactions • Seismology • Fluid Mechanics • Geophysical Fluid Dynamics • Atmospheric Thermodynamics • Dynamic Meteorology • Climate Dynamics & Climate Change • Physical Oceanography • Analysis of Physical Oceanographic Data • Marine Chemistry • Statistical Learning for Big Data • Artificial Neural Networks • Data Science for Engineers & Scientists • Simulation of Complex Systems • Stochastic Optimization Algorithms • Dynamical Systems • Computational Physics • Information Theory • Computational Biology • Systems Biology • Leadership for Sustainability Transitions • Environmental Risk Assessment • Science Writing

Undergraduate: Ice and the Climate System • Observational Methods in Physical Oceanography • Natural Disasters • Global Tectonics • Paleoclimatology • Earth Surface Processes • Organic Chemistry • Multivariate Calculus • Linear Algebra & Matrix Theory • Real Analysis • Vector Analysis and Differential Geometry • Combinatorics & Graph Theory • Algebraic Topology & Knot Theory • Probability Theory & Statistics • Econometrics • General Equilibrium Theory • Mathematical Game Theory • Micro- & Macroeconomics • World Finance • Conservation Biology • Global Environmental Governance